

What the study found

LFS, RMFS, and a model that lost its meaning structure — Waiz Khan, JHU, August 2026

A degraded model scored best. Across 25 training variants of Qwen3-0.6B (75 checkpoints), the variant trained with the word level alignment objective lost about **95%** of the structure in its representations that encodes meaning rather than language. It posted the **best LFS score in the study**. MEXA ranked it third of the 25.

The ratio cannot see this. LFS is a share: meaning structure against language structure. When a model shrinks both together the share does not move. Synthetic tests give the closed form — the share shifts by less than a millionth while meaning structure falls by 99 percent. Any scale invariant ratio inherits this. The possibility was written into the July preregistration, together with the detector that catches it, before any of these models were trained.

The damage cost it nothing. On a 40 language reading test the degraded variant scores **0.395** against **0.390** for the untouched control. The variants that did lose ability, by roughly six points, look clean on every geometric measure taken. Representational damage and capability damage are separable, and the measures that see one do not see the other.

RMFS reports four readings per language. Each carries its own validated number, and each predicts something the others miss:

reading	what it tracks	correlation
Meaning against language	carrying content across languages	0.76
Sentence matching	downstream exam performance	0.24–0.31
Weakest aligned languages	which variants lost ability	0.51
Collapse flag	the case above	exact

The first is LFS’s original measurement, kept intact. Correlations are measured after removing the effects of training data volume, language family, script and tokenizer differences.

Scores read as percentiles. For a model: embed 300 parallel sentences in 41 languages at one layer, compute the four readings per language, convert each to a percentile against a reference grid of 14 models, average. A score of 74 means better than 74 percent of the reference models. Roughly four GPU minutes per model. The ordering falls out with no tuning: Qwen3-8B **90.5**, 4B **85.7**, 1.7B **73.8**; the multilingual focused Salamandra-7B **69.0**; the English centric OLMo-2-1B **28.6**.

What the mapping between languages looks like. Across 1,792 model and language pairs the fitted map is linear or simpler in **74 percent** of cases. The nonlinear cases are concentrated rather than spread out: three models carry 80 percent of them, and across the other eleven only 6.5 percent of languages need a nonlinear map.